Администрирование сетевых подсистем

Базовая защита от атак типа «brute force» с использованием Fail2ban

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Цели и задачи работы

Цель лабораторной работы

Получить практические навыки настройки и использования Fail2ban
для базовой защиты сервера от атак типа «brute force»
на примере служб SSH, HTTP и почтовых сервисов.

Выполнение лабораторной работы

Установка и запуск Fail2ban

```
[akabrikosov@server.akabrikosov.net sambashare]$ sudo -i tail -f /var/log/fail2ban.log
[sudo] password for akabrikosov:
2026-01-10 09:12:31,764 fail2ban.server        [24028]: INFO    ----------------------------------
--------------
2026-01-10 09:12:31,764 fail2ban.server        [24028]: INFO    Starting Fail2ban v1.1.0
2026-01-10 09:12:31,764 fail2ban.observer      [24028]: INFO    Observer start...
2026-01-10 09:12:31,767 fail2ban.database      [24028]: INFO    Connected to fail2ban persistent dat
abase '/var/lib/fail2ban/fail2ban.sqlite3'
2026-01-10 09:12:31,768 fail2ban.database      [24028]: WARNING New database created. Version '4'
```

Рис. 1: Просмотр журнала Fail2ban после запуска сервиса

Настройка защиты SSH

Open

customisation.local
/etc/fail2ban/jail.d

```
1 [DEFAULT]
2 bantime = 3600
3
4 [sshd]
5 port = ssh,2022
6 enabled = true
7
8 [sshd-ddos]
9 filter = sshd
10 enabled = true
11
12 [selinux-ssh]
13 enabled = true
```

Проверка запуска jail SSH

```
2026-01-10 09:16:21,502 fail2ban.actions        [24769]: INFO      banTime: 3600
2026-01-10 09:16:21,502 fail2ban.filter         [24769]: INFO      encoding: UTF-8
2026-01-10 09:16:21,503 fail2ban.filter         [24769]: INFO    Added logfile: '/var/log/audit/audit
.log' (pos = 0, hash = 96ade6aee80cad1cffb8aa95c4d6cb793fb74bb2)
2026-01-10 09:16:21,504 fail2ban.jail           [24769]: INFO    Creating new jail 'sshd-ddos'
2026-01-10 09:16:21,504 fail2ban.jail           [24769]: INFO    Jail 'sshd-ddos' uses pyinotify {}
2026-01-10 09:16:21,504 fail2ban.jail           [24769]: INFO    Initiated 'pyinotify' backend
2026-01-10 09:16:21,505 fail2ban.filter         [24769]: INFO      maxLines: 1
2026-01-10 09:16:21,505 fail2ban.filter         [24769]: INFO      maxRetry: 5
2026-01-10 09:16:21,505 fail2ban.filter         [24769]: INFO      findtime: 600
2026-01-10 09:16:21,505 fail2ban.actions        [24769]: INFO      banTime: 3600
2026-01-10 09:16:21,505 fail2ban.filter         [24769]: INFO      encoding: UTF-8
2026-01-10 09:16:21,505 fail2ban.jail           [24769]: INFO    Jail 'sshd' started
2026-01-10 09:16:21,506 fail2ban.jail           [24769]: INFO    Jail 'selinux-ssh' started
2026-01-10 09:16:21,507 fail2ban.jail           [24769]: INFO    Jail 'sshd-ddos' started
2026-01-10 09:16:21,508 fail2ban.filter         [24769]: WARNING [selinux-ssh] Ignoring all log entri
es older than 600s; these are probably messages generated while fail2ban was not running.
2026-01-10 09:16:21,508 fail2ban.filter         [24769]: WARNING [selinux-ssh] Please check a jail fo
r a timing issue. Line with odd timestamp: type=BPF msg=audit(1768035976.987:8388): prog-id=2516 op=U
NLOAD
2026-01-10 09:16:21,510 fail2ban.filtersystemd  [24769]: INFO    [sshd] Jail is in operation now (pro
cess new journal entries)
```

Рис. 3: Журнал Fail2ban после включения защиты SSH

Настройка защиты веб-служб (Apache)

Open

customis
/etc/fail

```
18 [apache-badbots]
19 enabled = true
20
21 [apache-noscript]
22 enabled = true
23
24 [apache-overflows]
25 enabled = true
26
27 [apache-nohome]
28 enabled = true
29
30 [apache-botsearch]
31 enabled = true
32
33 [apache-fakegooglebot]
34 enabled = true
35
36 [apache-modsecurity]
37 enabled = true
38
39 [apache-shellshock]
40 enabled = true
41
```

Проверка защиты HTTP

```
2026-01-10 09:18:44,760 fail2ban.jail           [25116]: INFO    Creating new jail 'sshd-ddos'
2026-01-10 09:18:44,760 fail2ban.jail           [25116]: INFO    Jail 'sshd-ddos' uses pyinotify {}
2026-01-10 09:18:44,761 fail2ban.jail           [25116]: INFO    Initiated 'pyinotify' backend
2026-01-10 09:18:44,761 fail2ban.filter         [25116]: INFO      maxLines: 1
2026-01-10 09:18:44,762 fail2ban.filter         [25116]: INFO      maxRetry: 5
2026-01-10 09:18:44,762 fail2ban.filter         [25116]: INFO      findtime: 600
2026-01-10 09:18:44,762 fail2ban.actions        [25116]: INFO      banTime: 3600
2026-01-10 09:18:44,762 fail2ban.filter         [25116]: INFO      encoding: UTF-8
2026-01-10 09:18:44,762 fail2ban.filtersystemd  [25116]: INFO    [sshd] Jail is in operation now (pro
cess new journal entries)
2026-01-10 09:18:44,762 fail2ban.jail           [25116]: INFO    Jail 'sshd' started
2026-01-10 09:18:44,762 fail2ban.jail           [25116]: INFO    Jail 'selinux-ssh' started
2026-01-10 09:18:44,764 fail2ban.jail           [25116]: INFO    Jail 'apache-auth' started
2026-01-10 09:18:44,766 fail2ban.jail           [25116]: INFO    Jail 'apache-badbots' started
2026-01-10 09:18:44,766 fail2ban.jail           [25116]: INFO    Jail 'apache-noscript' started
2026-01-10 09:18:44,767 fail2ban.jail           [25116]: INFO    Jail 'apache-overflows' started
2026-01-10 09:18:44,767 fail2ban.jail           [25116]: INFO    Jail 'apache-nohome' started
2026-01-10 09:18:44,768 fail2ban.jail           [25116]: INFO    Jail 'apache-botsearch' started
2026-01-10 09:18:44,768 fail2ban.jail           [25116]: INFO    Jail 'apache-fakegooglebot' started
2026-01-10 09:18:44,770 fail2ban.jail           [25116]: INFO    Jail 'apache-modsecurity' started
2026-01-10 09:18:44,770 fail2ban.jail           [25116]: INFO    Jail 'apache-shellshock' started
2026-01-10 09:18:44,771 fail2ban.jail           [25116]: INFO    Jail 'sshd-ddos' started
```

Рис. 5: Журнал Fail2ban после включения защиты HTTP

Настройка защиты почтовых сервисов

Open customi /etc/fa

```
29
30 [apache-botsearch]
31 enabled = true
32
33 [apache-fakegooglebot]
34 enabled = true
35
36 [apache-modsecurity]
37 enabled = true
38
39 [apache-shellshock]
40 enabled = true
41
42 [postfix]
43 enabled = true
44
45 [postfix-rbl]
46 enabled = true
47
48 [dovecot]
49 enabled = true
50
51 [postfix-sasl]
52 enabled = true
```

Проверка защиты почты

```
cess new journal entries)
2026-01-10 09:21:00,693 fail2ban.jail           [25457]: INFO    Jail 'sshd' started
2026-01-10 09:21:00,693 fail2ban.jail           [25457]: INFO    Jail 'selinux-ssh' started
2026-01-10 09:21:00,694 fail2ban.jail           [25457]: INFO    Jail 'apache-auth' started
2026-01-10 09:21:00,694 fail2ban.jail           [25457]: INFO    Jail 'apache-badbots' started
2026-01-10 09:21:00,694 fail2ban.jail           [25457]: INFO    Jail 'apache-noscript' started
2026-01-10 09:21:00,695 fail2ban.jail           [25457]: INFO    Jail 'apache-overflows' started
2026-01-10 09:21:00,695 fail2ban.jail           [25457]: INFO    Jail 'apache-nohome' started
2026-01-10 09:21:00,695 fail2ban.jail           [25457]: INFO    Jail 'apache-botsearch' started
2026-01-10 09:21:00,696 fail2ban.jail           [25457]: INFO    Jail 'apache-fakegooglebot' started
2026-01-10 09:21:00,696 fail2ban.jail           [25457]: INFO    Jail 'apache-modsecurity' started
2026-01-10 09:21:00,697 fail2ban.jail           [25457]: INFO    Jail 'apache-shellshock' started
2026-01-10 09:21:00,697 fail2ban.jail           [25457]: INFO    Jail 'postfix' started
2026-01-10 09:21:00,698 fail2ban.jail           [25457]: INFO    Jail 'postfix-rbl' started
2026-01-10 09:21:00,698 fail2ban.jail           [25457]: INFO    Jail 'dovecot' started
2026-01-10 09:21:00,699 fail2ban.jail           [25457]: INFO    Jail 'postfix-sasl' started
2026-01-10 09:21:00,700 fail2ban.jail           [25457]: INFO    Jail 'sshd-ddos' started
2026-01-10 09:21:00,701 fail2ban.filtersystemd  [25457]: INFO    [dovecot] Jail is in operation now (
process new journal entries)
2026-01-10 09:21:00,701 fail2ban.filtersystemd  [25457]: INFO    [postfix-rbl] Jail is in operation n
ow (process new journal entries)
2026-01-10 09:21:00,701 fail2ban.filtersystemd  [25457]: INFO    [postfix] Jail is in operation now (
process new journal entries)
2026-01-10 09:21:00,701 fail2ban.filtersystemd  [25457]: INFO    [postfix-sasl] Jail is in operation
```

Рис. 7: Журнал Fail2ban после включения защиты почты

Проверка статуса Fail2ban

```
[root@server.akabrikosov.net server]#
[root@server.akabrikosov.net server]# fail2ban-client status
Status
|- Number of jail:      16
`- Jail list:   apache-auth, apache-badbots, apache-botsearch, apache-fakegooglebot, apache-modsecuri
ty, apache-nohome, apache-noscript, apache-overflows, apache-shellshock, dovecot, postfix, postfix-rb
l, postfix-sasl, selinux-ssh, sshd, sshd-ddos
[root@server.akabrikosov.net server]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 0
|  |- Total failed:     0
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:     0
   `- Banned IP list:
[root@server.akabrikosov.net server]# fail2ban-client set sshd maxretry 2
2
[root@server.akabrikosov.net server]#
```

Рис. 8: Проверка статуса Fail2ban

Блокировка IP-адреса клиента

```
[root@server.akabrikosov.net server]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 1
|  |- Total failed:     3
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 1
   |- Total banned:     1
   `- Banned IP list:   192.168.1.30
[root@server.akabrikosov.net server]# fail2ban-client set sshd unbanip 192.168.1.30
1
[root@server.akabrikosov.net server]# fail2ban-client status sshd
Status for the jail: sshd
|- Filter
|  |- Currently failed: 1
|  |- Total failed:     3
|  `- Journal matches:  _SYSTEMD_UNIT=sshd.service + _COMM=sshd + _COMM=sshd-session
`- Actions
   |- Currently banned: 0
   |- Total banned:     1
   `- Banned IP list:
[root@server.akabrikosov.net server]#
```

Рис. 9: Блокировка IP-адреса клиента

Разблокировка и исключение IP-адреса

Open

customisation.lo
/etc/fail2ban/jail.c

```
1 [DEFAULT]
2 bantime = 3600
3 ignoreip = 127.0.0.1/8 192.168.1.30
4
5 [sshd]
6 port = ssh,2022
7 enabled = true
8
9 [sshd-ddos]
10 filter = sshd
11 enabled = true
12
13 [selinux-ssh]
14 enabled = true
15
16 [apache-auth]
17 enabled = true
18
19 [apache-badbots]
20 enabled = true
```

Проверка ignoreip

```
2026-01-10 09:26:19,387 fail2ban.filter         [26199]: INFO    [sshd] Ignore 192.168.1.30 by ip
2026-01-10 09:26:23,367 fail2ban.filter         [26199]: INFO    [sshd] Ignore 192.168.1.30 by ip
2026-01-10 09:26:26,526 fail2ban.filter         [26199]: INFO    [sshd] Ignore 192.168.1.30 by ip
```

Рис. 11: Журнал Fail2ban с ignoreip

Подготовка provisioning-конфигурации

Vagrantfile | smb.sh | smb.sh | netlog.sh | netlog.sh | protect.sh

```
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y install fail2ban
echo "Copy configuration files"
cp -R /vagrant/provision/server/protect/etc/* /etc
restorecon -vR /etc
echo "Start fail2ban service"
systemctl enable fail2ban
systemctl start fail2ban
```

Рис. 12: Скрипт protect.sh

Выводы по проделанной работе

Вывод

В ходе лабораторной работы была реализована базовая защита сервера от атак типа «brute force» с использованием Fail2ban.

Настроена защита SSH, веб-служб и почтовых сервисов, проверены механизмы блокировки, разблокировки и исключений IP-адресов.

Дополнительно выполнена автоматизация настройки Fail2ban в процессе развертывания виртуальной машины с помощью Vagrant.